Saturation Excitation in Infinite Dimensions with Application to Neural Field Kernel Identification ⋆

Adel M. Annabi [a], Jean-Baptiste Pomet [a], Dario Prandi [b], Ludovic Sacchelli [a]

[a] *Université Côte d'Azur, Inria, CNRS, LJAD, France*

[b] *Université Paris-Saclay, CNRS, CentraleSupélec, Laboratoire des Signaux et Systèmes, 91190, Gif-sur-Yvette, France*

Abstract

We introduce an excitation design for identifying operators that act on the saturation of a function-valued state. The design prescribes known saturation profiles on successive time intervals and determines the operator on their finite-dimensional span. Partition refinement yields operator-norm convergence for compact operators and a quantitative error bound under spatial regularity. For a neural field with measured voltage and unknown connectivity kernel, we construct distributed inputs that realize the required indicator profiles uniformly over the admissible kernels and initial states. A weighted integral of the measured voltage and known input recovers each operator action without differentiating the data. The estimator processes measurements causally and updates the kernel at the end of each probing window. For kernels with a square-integrable derivative in the input variable, every prescribed accuracy is attained after a known finite horizon under ideal measurements. We give a deterministic measurement-error bound and discuss smooth spatial stimulation.

Key words: Identification; excitation design; persistency of excitation; infinite-dimensional systems; neural fields; saturation; kernel estimation.

1 Introduction

Identification depends on excitation: the signals multiplying unknown parameters must distinguish their possible values. In nonlinear systems these signals are generated by the state, so designing an informative input can itself depend on uncertain dynamics. Saturation provides a different way to prescribe the regressor. Once a trajectory enters a region where the activation is constant, the regressor takes a known value even if the state continues to evolve. The identification question is then whether inputs can produce enough prescribed saturation values, in a known order and on known intervals.

We extend the saturation-excitation construction to function-valued states and unknown operators. In finite dimension, a finite set of independent saturation values can identify an entire matrix. In infinite dimension, finitely many probing profiles span only a finite-dimensional subspace. The excitation design must therefore specify an approximation space as well as a sequence of time windows. We use indicators of partition cells: each experiment returns one operator action, and refinement makes the unprobed remainder small for compact operators. Regularity of an integral kernel gives an explicit accuracy as a function of the partition width.

The construction is realized on a neural field with distributed input, unknown connectivity, and measurement of the voltage field. The same stimulation amplitude and settling-time bound work for every cell and every admissible kernel. These bounds allow the probing windows to be scheduled beforehand. On each window the recurrent current is constant, although the voltage need not be at equilibrium. A weighted integral of the voltage and the known input then recovers the operator action exactly in the ideal observation model. Measurements are processed as they arrive, and the kernel estimate is updated after each window.

Neural fields describe nonlocal interactions between neuronal populations [1]; identification from data is studied in [6]. Adaptive operator observers estimate connectivity under detectability or persistency-of-excitation conditions [4,5]. Here the informative profiles are explicitly generated by the control, and an approximation guar-

⋆ Corresponding author: Adel M. Annabi.

Email addresses: adel-malik.annabi@inria.fr (Adel M. Annabi), jean-baptiste.pomet@inria.fr (Jean-Baptiste Pomet), dario.prandi@centralesupelec.fr (Dario Prandi), ludovic.sacchelli@inria.fr (Ludovic Sacchelli).

antee is available at a finite completion time. This requires signed distributed actuation and a known current bound. Since the voltage is measured, its activation is computable; the contribution is to prescribe the profiles and certify their realization and the resulting approximation.

Weighted integral regression is an established way to avoid differentiating measured data [2]. We use it to extract the action of the unknown operator once saturation has fixed the regressor. The excitation construction, rather than the integration-by-parts identity, supplies the identification mechanism. Exact plateaus and indicator inputs enter the theorems; practical smooth approximations are discussed separately.

2 Saturation excitation and operator recovery

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain, $\mathcal{H} = L^2(\Omega;\mathbb{R})$, and $\mathcal{G}$ a real Hilbert space. Norms and inner products refer to these spaces, and $\|\cdot\|_{\mathrm{op}}$ denotes the operator norm. Consider an unknown bounded linear operator $W : \mathcal{H} \to \mathcal{G}$ and the measured response

$$r(t) = WS(X(t)), \qquad X \in C([0,T];\mathcal{H}). \tag{1}$$

The known, globally Lipschitz activation $S : \mathbb{R} \to [0,1]$ acts pointwise and has thresholds $\underline{R} < \overline{R}$ such that

$$S(v) = 0 \ (v \le \underline{R}), \qquad S(v) = 1 \ (v \ge \overline{R}). \tag{2}$$

Spatial identities hold almost everywhere; signal integrals are Bochner integrals.

Definition 1. Let $(Q_k)_{k=1}^N$ be a finite measurable partition of Ω with positive cell measures, and let $J_k \subset [0,T]$ be ordered observation intervals of lengths $\tau_k > 0$ with disjoint interiors. The trajectory X is saturation-exciting for this partition if

$$S(X(t)) = \mathbf{1}_{Q_k}, \qquad t \in J_k, \quad k = 1,\dots,N.$$

Set $e_k = \mathbf{1}_{Q_k}/\sqrt{|Q_k|}$ and $E = \operatorname{span}\{e_1,\dots,e_N\}$. These functions are orthonormal; write P_E for the orthogonal projection onto E. Exact excitation has a direct information consequence. For $f = \sum_j a_j e_j$ and $t \in J_k$, $\langle f, S(X(t))\rangle = a_k\sqrt{|Q_k|}$. Hence

$$\sum_{k=1}^{N} \frac{1}{\tau_k|Q_k|}\int_{J_k} |\langle f, S(X(t))\rangle|^2\,dt = \|f\|^2, \qquad f \in E. \tag{3}$$

The normalized finite-window Gramian on E is the identity. This is the finite-dimensional excitation that the experiment creates. It does not assert excitation of directions outside E: on the whole space, no square-integrable signal can satisfy such a lower bound with a positive constant, because its finite-window Gramian is compact [7]. Refinement and compactness of W control the unprobed remainder. If the profiles are only approximate, the stability estimate below quantifies the loss. From the responses define

$$\begin{aligned} z_k &= \frac{1}{\tau_k}\int_{J_k} r(t)\,dt, \\ \widehat{W}_E f &= \sum_k \frac{\langle f, \mathbf{1}_{Q_k}\rangle}{|Q_k|} z_k. \end{aligned} \tag{4}$$

For a sequence of partitions, let $h = \max_k \operatorname{diam}(Q_k) \to 0$ and denote the projections by P_h.

Theorem 2. *Saturation excitation gives $\widehat{W}_E = WP_E$. If W is compact, the estimates on refining partitions converge to W in operator norm. If, in addition,*

$$W^* \in \mathcal{L}(\mathcal{G}, H^1(\Omega)) \tag{5}$$

and the cells satisfy the uniform Poincaré estimate

$$\|f - f_{Q_k}\|_{L^2(Q_k)} \le C_P h \|\nabla f\|_{L^2(Q_k)}, \tag{6}$$

for $f \in H^1(\Omega)$, where $f_{Q_k} = |Q_k|^{-1}\int_{Q_k} f$ and C_P is independent of the partition, then

$$\|W - \widehat{W}_h\|_{\mathrm{op}} \le C_P h \|\nabla W^*\|_{\mathrm{op}}. \tag{7}$$

Proof. Exact excitation gives $z_k = W\mathbf{1}_{Q_k}$, so substitution in (4) yields WP_E. Cell averaging converges to the identity on $C(\overline{\Omega})$ by uniform continuity as $h \to 0$, and hence on $\mathcal{H}$ by density and $\|P_h\|_{\mathrm{op}} = 1$. The convergence is uniform on compact subsets by a finite covering argument. Apply this to the compact closure of the image under W^* of the unit ball of $\mathcal{G}$ to obtain

$$\|W(I - P_h)\|_{\mathrm{op}} = \|(I - P_h)W^*\|_{\mathrm{op}} \longrightarrow 0.$$

Finally, square (6) and sum over the cells with $f = W^*g$. Taking the supremum over $\|g\| \le 1$ proves (7). □

The mesh condition (6) is the usual approximation assumption for cellwise constants [3]; it holds, for example, on uniformly shape-regular simplicial meshes. It is required for the rate, not for exact recovery. Likewise, (5) is a sufficient regularity condition rather than a prerequisite for identification. If $W^*(\mathcal{G}) \subset H^1(\Omega)$, boundedness into H^1 follows from the closed graph theorem. The compact embedding $H^1(\Omega) \hookrightarrow L^2(\Omega)$ then already implies compactness of W. Without compactness, operator-norm convergence cannot hold for finite-rank estimates; for example, $\|I - P_h\|_{\mathrm{op}} = 1$.

2.1 Measurement noise and stability

The responses in (1) were ideal. In a realized experiment, the available signal is

$$\widetilde{r}(t) = WS(X(t)) + n(t), \tag{8}$$

where $n(t) \in \mathcal{G}$ is a measurement perturbation. Replacing r by $\widetilde{r}$ in (4) gives

$$\begin{aligned} \widetilde{z}_k &= \frac{1}{\tau_k}\int_{J_k} \widetilde{r}(t)\,dt = W\mathbf{1}_{Q_k} + \nu_k, \\ \nu_k &= \frac{1}{\tau_k}\int_{J_k} n(t)\,dt. \end{aligned} \tag{9}$$

Thus ν_k is the perturbation propagated through the window average; it is not an additional model parameter. Let $\widetilde{W}_E$ be the estimate obtained from $\widetilde{z}_k$.

Theorem 3. *For arbitrary response perturbations $\nu_k \in \mathcal{G}$,*

$$\|\widetilde{W}_E - WP_E\|_{\mathrm{op}} \le \left(\sum_k \frac{\|\nu_k\|^2}{|Q_k|}\right)^{1/2}. \tag{10}$$

Thus the total error is bounded by this quantity plus $\|W(I-P_E)\|_{\mathrm{op}}$. For compact W, convergence follows whenever the mesh and the right-hand side of (10) *tend to zero.*

Proof. The error applied to f is $\sum_k \langle f, e_k\rangle \nu_k/\sqrt{|Q_k|}$. Cauchy–Schwarz and Bessel's inequality give (10); the remaining claims follow by the triangle inequality and Theorem 2. □

If $\sup_{t\in J_k}\|n(t)\| \le \bar{n}$, then $\|\nu_k\| \le \bar{n}$ and

$$\|\widetilde{W}_E - WP_E\|_{\mathrm{op}} \le \bar{n}\left(\sum_{k=1}^{N}\frac{1}{|Q_k|}\right)^{1/2}. \tag{11}$$

For equal cells of measure h^d, the right-hand side is $\bar{n}h^{-d}$. This is a deterministic worst-case bound; it does not assume independence, zero mean, or cancellation between windows.

The same decomposition covers imperfect profiles. If $S(X(t)) = \mathbf{1}_{Q_k} + d_k(t)$ on J_k, then

$$\widetilde{z}_k - W\mathbf{1}_{Q_k} = W\left(\frac{1}{\tau_k}\int_{J_k} d_k(t)\,dt\right) + \nu_k. \tag{12}$$

The profile contribution is at most $\|W\|_{\mathrm{op}}\delta_k$, where $\delta_k = \sup_{t\in J_k}\|d_k(t)\|$. Measurement and profile errors therefore enter through the same normalized sum (10); the triangle inequality combines them. The voltage implementation below specializes (9) to the weighted integral and gives the corresponding factor $1 + 3/\tau$.

3 Neural field kernel estimation

Let $\Omega \subset \mathbb{R}^d$ be the bounded Lipschitz domain of Section 2, $\mathcal{H} = L^2(\Omega;\mathbb{R})$, and $\mathcal{G} = \mathcal{H}$. The unknown operator is

$$\begin{aligned} (Wf)(x) &= \int_\Omega K(x,s)f(s)\,ds, \\ K &\in L^2(\Omega\times\Omega). \end{aligned} \tag{13}$$

It is compact. The voltage field and its measurement satisfy

$$\dot{V} = -V + WS(V) + u, \qquad y = V, \tag{14}$$

with unit leakage after time scaling. In the notation of Section 2, $X(t) = V(\cdot,t)$. For $u \in L^1_{\mathrm{loc}}([0,\infty);\mathcal{H})$, global Lipschitz continuity of $-V + WS(V)$ gives a unique global mild solution, continuous and locally absolutely continuous in $\mathcal{H}$ [9]. It is C^1 between switches for piecewise continuous controls, including piecewise affine controls. Our experiment uses piecewise constant stages. We assume a calibrated full-field voltage measurement, motivated by voltage-sensitive imaging [8]; finite sensor averaging is outside the present analysis.

Assume $\|V_0\|_\infty \le \rho$ and a known row bound

$$\operatorname*{ess\,sup}_{x\in\Omega}\int_\Omega |K(x,s)|\,ds \le b. \tag{15}$$

Then $\|WS(v)\|_\infty \le b$. These bounds determine the input design. For accuracy rates, the sufficient kernel condition $K \in L^2(\Omega_x; H^1(\Omega_s))$ implies

$$\|\partial_s W^* g\| \le \|\partial_s K\|_{L^2}\|g\|.$$

It therefore supplies (5), as well as an L^2-kernel rate below. Adjoint regularity alone is an operator condition and need not give this stronger kernel regularity.

3.1 Producing the indicator profiles

Fix a partition $(Q_k)_{k=1}^N$ and an observation duration $\tau > 0$. Put $R = \max\{|\underline{R}|, |\overline{R}|\}$ and choose

$$\begin{aligned} &\mu > b + R, \qquad B = \max\{\rho, b+\mu\}, \\ &T_s \ge \log\frac{B+\mu-b}{\mu-b-R}. \end{aligned} \tag{16}$$

Set $L = T_s + \tau$, $s_k = (k-1)L$, and $J_k = [s_k + T_s, s_k + L]$. Apply

$$u(t) = \mu(2\mathbf{1}_{Q_k} - 1), \qquad s_k \le t < s_k + L, \tag{17}$$

and set $u = 0$ after $T = NL$.

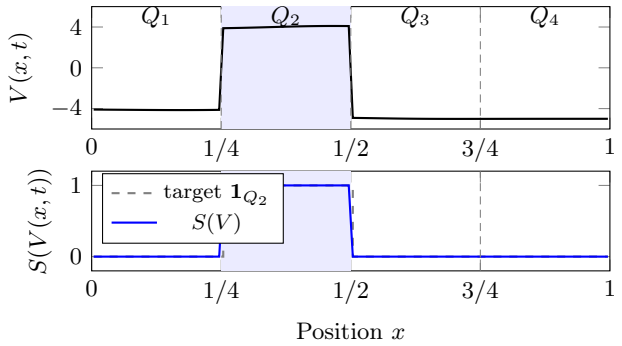


Figure 1. One-dimensional illustration of the construction: voltage and activation at the start of the second observation window of a four-cell experiment ($N = 4$). Shading marks the probed cell Q_2 and dashed lines the cell boundaries. The activation equals the target indicator on the $m = 128$ midpoint grid while the voltage keeps evolving.

Lemma 4. *Under* (2), (13), *and* (15), *the solution driven by* (17) *satisfies* $\|V(t)\|_\infty \le B$ *and*

$$S(V(t)) = \mathbf{1}_{Q_k}, \qquad t \in J_k.$$

The same amplitude and settling time work for every cell and partition, uniformly over the admissible kernels and initial states.

Proof. Variation of constants and (15) give

$$\|V(t)\|_\infty \le e^{-t}\rho + (b+\mu)(1-e^{-t}) \le B.$$

The integral formula also gives absolutely continuous scalar trajectories for almost every x, so pointwise comparison applies. On stage k, with $q = t - s_k$, it yields, on Q_k,

$$V(x,t) \ge (\mu - b) - (B + \mu - b)e^{-q}.$$

On its complement, the upper bound is $V(x,t) \le -(\mu - b) + (B + \mu - b)e^{-q}$. For $q \ge T_s$ these bounds are at least R and at most $-R$, respectively. The plateaus give the required indicator, and continuity in $\mathcal{H}$ includes the window endpoints. The state bound remains valid after the input is switched off. □

No state reset is used between stages. The acquisition time is $T = (T_s + \tau)N$, while the amplitude and settling time are independent of N. Figure 1 illustrates the activation profile and the evolving voltage.

3.2 Recovering the kernel from voltage

The response $WS(V)$ is not measured directly. On $J_k = [c_k, c_k + \tau]$, set $\theta = (t - c_k)/\tau$ and

$$\phi_k(t) = \frac{6}{\tau}\,\theta(1-\theta). \tag{18}$$

This nonnegative weight has unit integral and vanishes at both endpoints. Multiplying (14) by ϕ_k and integrating by parts gives

$$\begin{aligned} z_k &= \int_{J_k} \big[(\phi_k - \dot\phi_k)y - \phi_k u\big]\,dt \\ &= W \int_{J_k} \phi_k(t) S(V(t))\,dt. \end{aligned} \tag{19}$$

Only the known weight is differentiated. The voltage need not be at equilibrium. Define

$$\widehat{K}(x,s) = \sum_{k=1}^{N} \frac{z_k(x)}{|Q_k|}\mathbf{1}_{Q_k}(s), \tag{20}$$

and denote its integral operator by $\widehat{W}_h$; its kernel is $\widehat{K}_h$.

Proposition 5. *Under Lemma 4, ideal voltage measurements give* $\widehat{W}_h = WP_h$. *For a refining sequence with* $h \to 0$, $\|K - \widehat{K}_h\|_{L^2} \to 0$. *If* $K \in L^2(\Omega_x; H^1(\Omega_s))$ *and the partitions satisfy* (6) *with constant* C_P, *then*

$$\begin{aligned} \|W - \widehat{W}_h\|_{\mathrm{op}} &\le \|K - \widehat{K}_h\|_{L^2} \\ &\le C_P h \|\partial_s K\|_{L^2}. \end{aligned} \tag{21}$$

Proof. Lemma 4 and (19) give $z_k = W\mathbf{1}_{Q_k}$, so Theorem 2 applies. Substitution into (20) shows that $\widehat{K}_h$ is the average of $K(x,\cdot)$ over each input cell. Cell averaging converges to $K(x,\cdot)$ in L^2 for almost every x, with squared error bounded by $\|K(x,\cdot)\|^2$. Dominated convergence gives kernel convergence. The cell Poincaré inequality, squared and integrated in x, gives the rate with constant C_P. The operator norm is bounded by the kernel's L^2 norm. □

In general, C_P depends on the domain and on the shape regularity of the partition. For a known $M_1 \ge \|\partial_s K\|_{L^2}$, choose the coarsest admissible partition satisfying $C_P h M_1 \le \varepsilon$. With fixed $M_1 > 0$, the acquisition time $T = (T_s + \tau)N$ grows as the number of cells increases; for uniform cells in dimension d, $N = h^{-d}$ and $T = O(\varepsilon^{-d})$ as $\varepsilon \to 0$.

For perturbed responses $\widetilde{z}_k$, let $\widetilde{K}_h$ be the corresponding kernel estimate. Theorem 3 gives the operator estimate, and the triangle inequality gives

$$\begin{aligned} \|K - \widetilde{K}_h\|_{L^2} \le \|K - \widehat{K}_h\|_{L^2} \\ + \left(\sum_k \frac{\|\widetilde{z}_k - W\mathbf{1}_{Q_k}\|^2}{|Q_k|}\right)^{1/2}. \end{aligned} \tag{22}$$

A measurable voltage error with $\sup_t \|n(t)\| \le \bar{n}$ contributes at most $(1+3/\tau)\bar{n}$ per response, since $\|\phi_k\|_{L^1} =$

1 and $\|\dot{\phi}_k\|_{L^1} = 3/\tau$. Its aggregate contribution is at most $(1+3/\tau)\bar{n}\, h^{-d}$ for uniform cells. This is a deterministic worst-case bound; it does not assume cancellation between errors.

Remark 6. On each stage, replace the ideal input by a spatially regularized input u_ℓ of magnitude at most μ, agreeing with it outside the boundary layer $A_k = \{s \in \Omega : \operatorname{dist}(s, \partial Q_k) \le \ell\}$, where $0 < \ell \le h$. The same state bound and comparison apply outside A_k, with the same settling time. Let $\widehat{K}_h^\ell$ be the estimate obtained with u_ℓ. Using the actual input in (19), put

$$d_k = \int_{J_k} \phi_k(t)[S(V(t)) - \mathbf{1}_{Q_k}]\, dt.$$

Then $|d_k(x)| \le 1$ and $d_k(x) = 0$ for $x \notin A_k$. Cauchy–Schwarz gives

$$|W d_k(x)|^2 \le |A_k| \int_{A_k} |K(x,s)|^2\, ds.$$

Assume the partition is shape regular, with $|A_k| \le C_b \ell h^{d-1}$ and boundary layers overlapping at most C_o times almost everywhere. Then

$$\begin{aligned} \|\widehat{K}_h^\ell - \widehat{K}_h\|_{L^2}^2 &= \sum_k \frac{\|W d_k\|^2}{|Q_k|} \\ &\le C_b C_o \frac{\ell}{h} \|K\|_{L^2}^2. \end{aligned} \tag{23}$$

Thus $\ell = o(h)$ suffices for kernel and operator convergence with ideal measurements. This is sufficient, not necessary. One construction is $u_\ell = \mu(2\eta_\ell * \mathbf{1}_{Q_k} - 1)$, extending indicators by zero and using a nonnegative unit-mass mollifier supported in $[-\ell, \ell]$. Spatial regularization leaves the temporal switching unchanged; Figure 2 illustrates the transition regions.

4 Numerical illustration

We now specialize to $d = 1$, $\Omega = (0,1)$, and uniform partitions $Q_k = [(k-1)h, kh)$ with $h = 1/N$. For these cells the Poincaré constant is $C_P = 1/\pi$. Consider the smooth signed kernel

$$\begin{aligned} K(x,s) = (1 + 0.3\cos(2\pi x)) \\ \times [1.4e^{-q(x,s)^2} - 0.9e^{-p(x,s)^2}], \end{aligned} \tag{24}$$

where $q(x,s) = (x - s - 0.12\sin(2\pi s))/0.12$ and $p(x,s) = (x - s + 0.08)/0.25$. Take $S(v) = \min\{1, \max\{0, v\}\}$ and $V(x,0) = 0.4\cos(2\pi x) + 0.2\sin(6\pi x)$. The bounds $\rho = 1$, $R = 1$, and $b = 3$ hold, since $|K| \le 1.3(1.4 + 0.9) < 3$. We use $\mu = 5$, $B = 8$, $T_s = \log 10$, and $\tau = 4$ at every resolution.

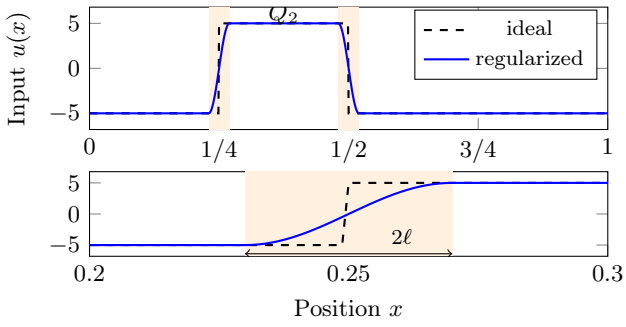


Figure 2. One-dimensional illustration of the regularized input during the probing of $Q_2 = [1/4, 1/2]$ on a four-cell partition, with support radius $\ell = 0.02$; the lower panel details the left cell boundary. Outside the ℓ-neighborhoods of the cell boundaries the profiles coincide, so saturation remains exact there; after the settling time, the activation mismatch is confined to those neighborhoods, as in Remark 6. The plotted cosine regularization is C^1 in space.

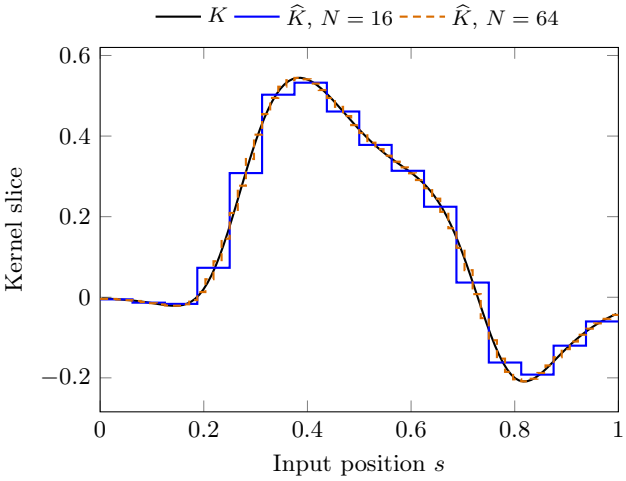


Figure 3. Kernel slice at $x = 129/256$: truth (black), reconstruction with $N = 16$ (blue), and with $N = 64$ (orange dashed). Both reconstructions use $m = 128$, $\Delta t_{\max} = 0.02$, and $\tau = 4$. Finer probing resolves more input-variable variation.

Midpoint quadrature on $m = 128$ uniform spatial points discretizes the integral operator. Fourth-order Runge–Kutta advances the voltage with maximum step 0.02, aligned with input switches and observation windows. Trapezoidal quadrature of (19) reconstructs the responses from voltage and the known input; kernel values are used only in the forward model and diagnostics. Each run uses N successive probing cells without resetting the voltage. Kernel errors use the two-dimensional midpoint rule on the simulation grid.

With ideal inputs, the activation equals the target at every checked window point. Figure 3 compares reconstructed slices for $N = 16$ and 64 at the same output location and numerical settings. Figure 4 shows the relative reconstruction error for $N = 2, 4, 8, 16, 32, 64$. Ta-

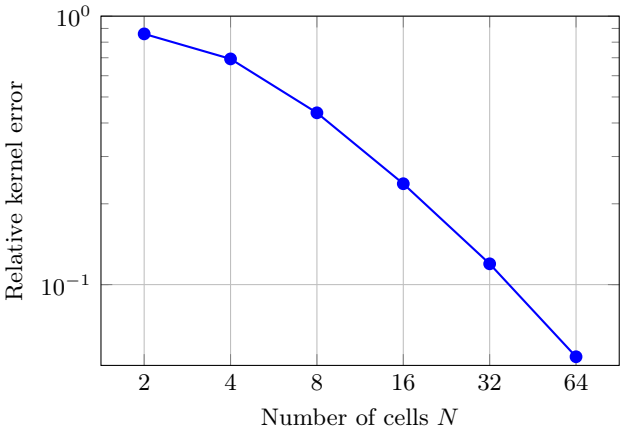


Figure 4. Relative reconstruction error $\|K - \widehat{K}_h\|_{L^2}/\|K\|_{L^2}$ versus the number of probing cells. All experiments use the same stimulation amplitude, stage duration, and observation window.

Table 1
Reconstruction error for $N = 2, 4, 8, 16, 32, 64$ probing cells. The absolute error is $\|K - \widehat{K}_h\|_{L^2(\Omega\times\Omega)}$; the relative error divides by $\|K\|_{L^2} = 0.3422$. All runs use $m = 128$, $\Delta t_{\max} = 0.02$, $\tau = 4$, and ideal measurements. The cell width is $h = 1/N$.

N	Abs. error	Rel. error
2	0.29396	0.85902
4	0.23717	0.69305
8	0.14936	0.43646
16	0.08130	0.23756
32	0.04090	0.11952
64	0.01843	0.05386

ble 1 reports the corresponding absolute and relative errors. The estimator uses only the measured voltage and the known input; the true kernel enters the simulation and the diagnostics, not the estimator. The acquisition time is $T = N(T_s + \tau)$, from 12.6 to 403.4 time units for $N = 2$ to 64.

For the noisy experiment, the primary perturbation is the bounded deterministic signal

$$n(x,t) = \frac{0.01}{1.5}\big[\sin(31t+2\pi x)+0.5\cos(17t-4\pi x)\big], \quad (25)$$

which satisfies $|n(x,t)| \le 0.01$ and does not assume statistical independence between samples. Figure 5 shows this trace and, for comparison only, one realization of zero-mean white Gaussian noise with standard deviation 0.01. The Gaussian perturbation is unbounded and is not covered by the deterministic bound. The estimator receives $y = V + n$, so the perturbation propagates through the weighted integral as in (9).

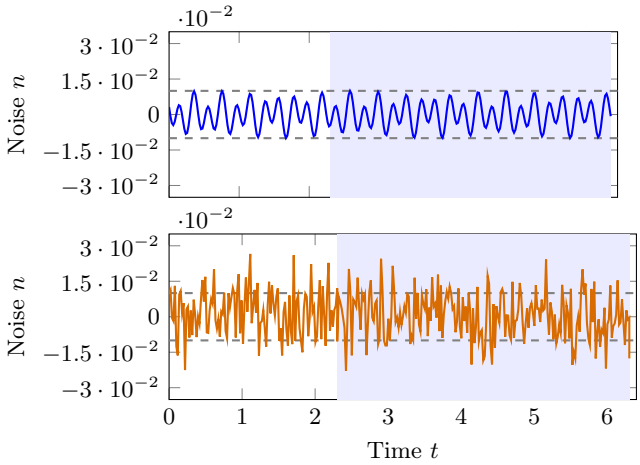


Figure 5. Noise traces at $x_0 = 129/256$ over the first stage with $\tau = 4$. Top: bounded sinusoidal perturbation, $|n| \le 0.01$. Bottom: one white Gaussian realization with standard deviation 0.01, comparison only; it is unbounded and not covered by the deterministic bound. Shaded: J_1; dashed: ± 0.01.

Figure 6 compares the reconstructed slice and pointwise error at $N = 64$ for the bounded sinusoidal and Gaussian perturbations. The bounded sinusoidal case gives relative L^2 errors 0.238 at $N = 16$ and 0.058 at $N = 64$. Over 20 Gaussian realizations, the relative errors are 0.2419 ± 0.0002 at $N = 16$ and 0.1920 ± 0.0016 at $N = 64$. The deterministic bounded test therefore stays close to the ideal error, while the Gaussian comparison shows the variance accumulation at fine resolution. The observation window controls the noise term.

For spatial regularization, we convolve each indicator with the unit-mass density $\eta_\ell(s) = \pi\cos(\pi s/(2\ell))/(4\ell)$ on $|s| \le \ell$, zero elsewhere. The resulting input is C^1 in space and satisfies Remark 6. At $N = 64$, with $m = 1024$ and $\Delta t_{\max} = 0.005$, the relative errors for $\ell/h = 0, 1/4, 1/2$ are 0.062, 0.063, and 0.095. These values are consistent with the predicted degradation in ℓ/h; they illustrate the effect of regularization at fixed probing resolution rather than an asymptotic rate.

5 Conclusion

Saturation excitation turns operator identification into a sequence of prescribed spatial probes. The finite experiment identifies the operator on a chosen subspace, and kernel regularity controls what that subspace leaves unprobed. For the measured neural field, distributed signed inputs realize the indicator profiles with uniform amplitude and timing bounds. Weighted integration extracts each operator action from incoming voltage measurements, without differentiating the data. Refinement improves reconstruction accuracy at the cost of a longer acquisition time; the observation window controls the resulting measurement-error tradeoff. Spatial regular-

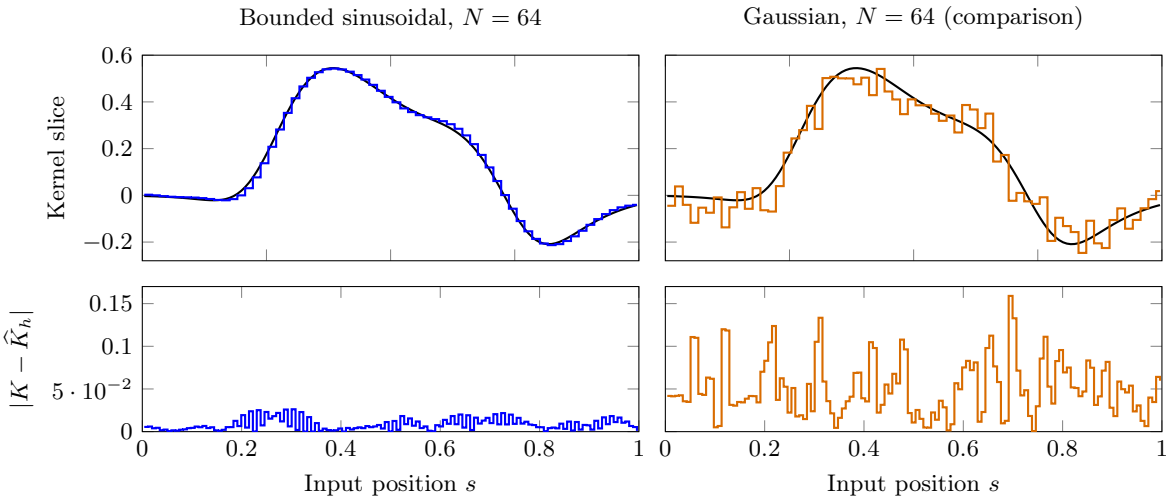


Figure 6. Kernel slice at $x_0 = 129/256$ for $N = 64$ with nominal noise 0.01. Top: truth K (black) and reconstruction $\widehat{K}_h$ (color). Bottom: pointwise absolute error. Left: bounded sinusoidal perturbation. Right: one realization of zero-mean white Gaussian noise with standard deviation 0.01, included for comparison only; it is unbounded and is not covered by the deterministic bound. The relative errors are 0.058 for the bounded sinusoidal case and 0.193 for this Gaussian realization.

ization preserves convergence when its transition width vanishes relative to the cell width. Extending the design to restricted actuators and spatially averaged measurements is a natural next step.

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